Experiment 8

Aim

To build and train an Artificial Neural Network (ANN) model to predict customer churn using a banking dataset.

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1. Dataset Source

Source:
https://www.kaggle.com/datasets/shrutimechlearn/churn-modelling

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2. Dataset Description

Records: ~10,000 customers

Features:

- Credit Score
- Geography
- Gender
- Age
- Tenure
- Balance
- Number of Products
- Has Credit Card
- Is Active Member
- Estimated Salary

Target Variable:

- `Exited` (0 = No churn, 1 = Churn)

Use Case: Predict whether a customer will leave the bank.

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3. Mathematical Formulation of the Algorithm

Neuron Output

$$y = f\left(\sum w_i x_i + b\right)$$

Activation Function (Sigmoid)

$$\sigma(x) = \frac{1}{1 + e^{-x}}$$

Loss Function (Binary Crossentropy)

$$L = -[y \log(\hat{y}) + (1 - y) \log(1 - \hat{y})]$$

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4. Algorithm Limitations

- Requires large data for good performance
- Prone to overfitting
- Requires tuning (layers, neurons, epochs)
- Black-box model (low interpretability)

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5. Methodology / Workflow

1. Data Collection
2. Preprocessing
3. Encoding
4. Scaling
5. ANN Model
6. Training

7. Evaluation
8. Prediction

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6. Performance Analysis

- Metric: Accuracy
- Higher accuracy = better prediction
- Can also use confusion matrix

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7. Hyperparameter Tuning

- Number of layers
- Number of neurons
- Learning rate
- Epochs
- Batch size

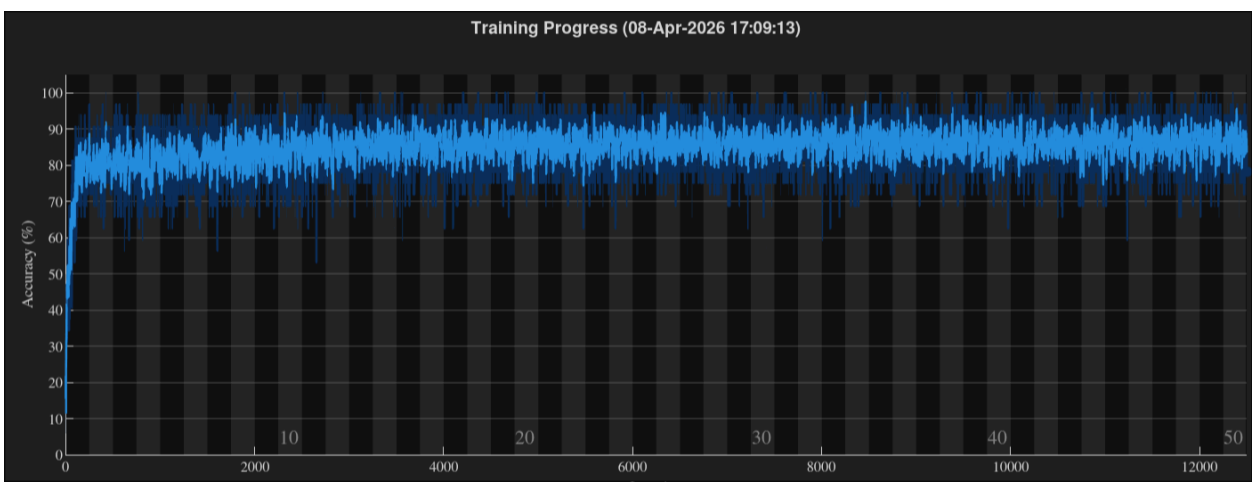
Code

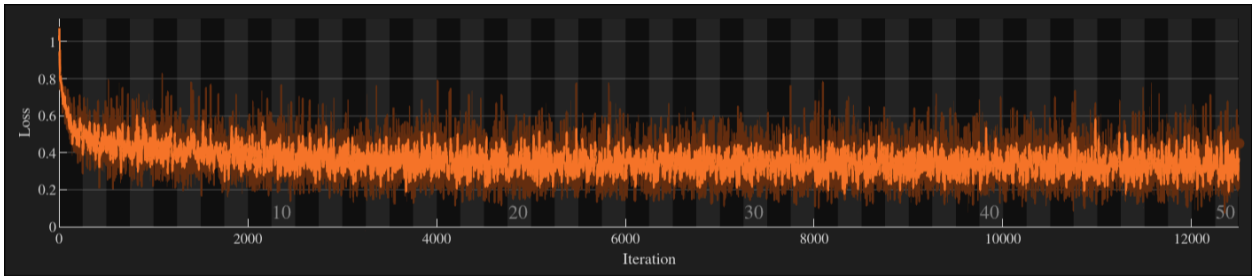
```
%% 1. Load Data
data = readtable('Churn_Modelling.csv');

%% 2. Preprocessing
data(:, {'RowNumber','CustomerId','Surname'}) = [];
data.Gender = double(strcmp(data.Gender, 'Male'));

geo = dummyvar(categorical(data.Geography));
geo = geo(:, 2:end);
data(:,'Geography') = [];
geoTable = array2table(geo, 'VariableNames',
{'Geography_Germany','Geography_Spain'});
data = [data, geoTable];
```

```
%% 3. Split X and y
y = double(data.Exited);
data(:,'Exited') = [];
X = table2array(data);

%% 4. Train-Test Split (80/20)
cv = cvpartition(length(y), 'HoldOut', 0.2);
X_train = X(training(cv), :);
X_test  = X(test(cv), :);
y_train = y(training(cv));
y_test  = y(test(cv));

%% 5. Standardize
[X_train, mu, sigma] = zscore(X_train);
X_test = (X_test - mu) ./ sigma;

%% 6. Build ANN
inputSize = size(X_train, 2);

layers = [
    featureInputLayer(inputSize)
    fullyConnectedLayer(6)
    reluLayer
    fullyConnectedLayer(6)
    reluLayer
    fullyConnectedLayer(2)        % 2 output neurons for binary
classification
    softmaxLayer
    classificationLayer
];

options = trainingOptions('adam', ...
    'MaxEpochs', 50, ...
    'MiniBatchSize', 32, ...
    'InitialLearnRate', 0.001, ...
```

```
    'Shuffle', 'every-epoch', ...
    'Verbose', true, ...
    'Plots', 'training-progress');

%% 7. Train
net = trainNetwork(X_train, categorical(y_train), layers,
options);

%% 8. Evaluate
y_pred = double(string(classify(net, X_test)));
accuracy = sum(y_pred == y_test) / length(y_test);
fprintf('\nTest Accuracy: %.2f%%\n', accuracy * 100);

%% 9. Confusion Matrix
figure;
confusionchart(y_test, y_pred, ...
    'Title', 'Confusion Matrix - Customer Churn ANN');
```

Output





Results

Validation accuracy:	N/A
Training finished:	Max epochs completed

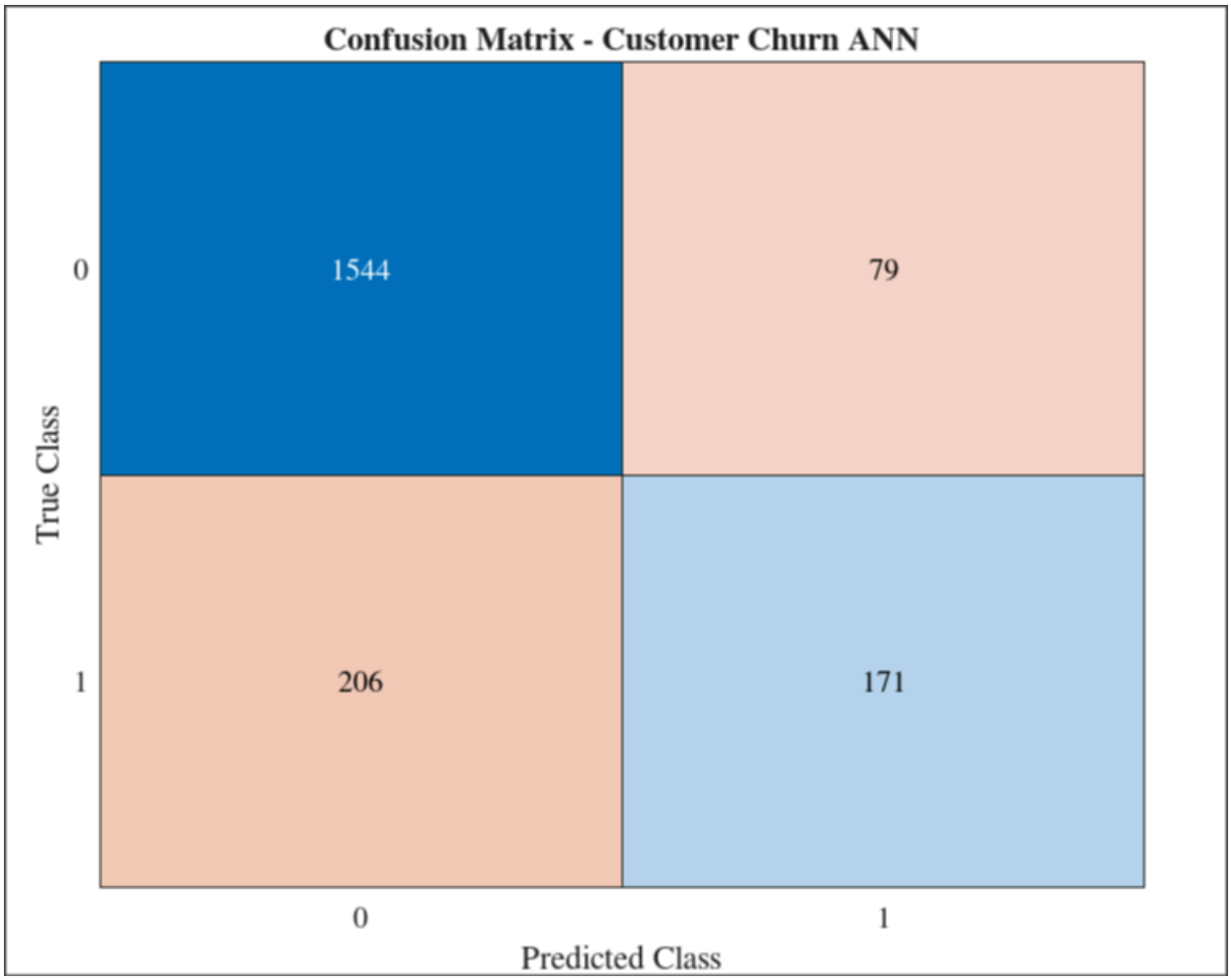
Training Time

Start time:	08-Apr-2026 17:09:13
Elapsed time:	3 min 9 sec

Training Cycle

Epoch:	50 of 50
Iteration:	12500 of 12500
Iterations per epoch:	250
Maximum iterations:	12500

```
Training finished: Max epochs completed.

Test Accuracy: 85.75%
>>
```



Conclusion

The ANN model successfully predicts customer churn with good accuracy. Neural networks are powerful for classification tasks but require proper tuning and preprocessing.